Gradient Inversion Attacks in Federated Learning

Deep Leakage from Gradients: Reconstructing Client Data from Shared Updates

October 15, 2026

Outline

1. Motivation & Threat Model
2. Attack Methods
3. DP-SGD Defense
4. Experimental Evaluation
5. Conclusion

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Federated Learning's Privacy Promise

K clients train a shared model without centralizing their data: each round, client k trains locally on private data $\mathcal{D}_k$ and returns only an update.

$$w_{t+1} = w_t + \sum_{k \in S_t} \frac{n_k}{\sum_{j \in S_t} n_j} \Delta w_k^{(t)}$$

- Only $\Delta w_k^{(t)}$ (or gradient $g_k^{(t)}$) is sent – **raw data is assumed to never leave the client**
- Server just aggregates updates (FedAvg); this is the privacy claim we test

The Gradient Inversion Problem

An **honest-but-curious** attacker (the server itself, or an eavesdropper) follows the protocol exactly, but tries to invert what it legitimately observes:

$$g = \nabla_w \, \ell\big(w_t;\, \{x_i, y_i\}_{i=1}^{B}\big) \quad \longrightarrow \quad \{(\hat{x}_i, \hat{y}_i)\}_{i=1}^{B}$$

- **Knows:** architecture, loss ℓ, weights w_t, leaked gradient g **Does not know:** the private batch
- Two answers to this problem: *optimize* a dummy batch to match g, or *solve* for it directly in closed form

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DLG: Deep Leakage from Gradients

The first attack to show a leaked gradient alone is enough: search for a dummy batch whose gradient matches the real one.

$$\min_{\hat{x},\hat{y}} \left\| g' - g \right\|_2^2, \qquad g' = \nabla_w \ell(w_t; \hat{x}, \hat{y})$$

- Dummy $(\hat{x}, \hat{y})$ start as random noise, jointly optimized by gradient descent – requires differentiating through g' itself ("double backprop")
- Slow and occasionally unstable, since image and label are searched together

iDLG: Closed-Form Label Recovery

Same gradient-matching idea as DLG, but the label $\hat{y}$ no longer needs to be searched for:

$$\hat{y} = \arg\min_{c} \left(\frac{\partial \ell}{\partial z_c} \right)$$

- For cross-entropy loss, the sign pattern of the last layer's gradient reveals the true label in one step
- Only the image $\hat{x}$ is then optimized against DLG's same objective – faster, more reliable convergence

Inverting Gradients

DLG's L_2 matching breaks down on deep nets, where gradient magnitude is dominated by whichever layer is largest. Inverting Gradients matches *direction* instead:

$$\min_{\hat{x}} \left(1 - \frac{\langle g', g \rangle}{\|g'\| \, \|g\|}\right) + \alpha \, \mathrm{TV}(\hat{x})$$

- Cosine cost + total-variation prior + sign-based Adam updates
- This is what lets the attack scale to ImageNet-sized ResNet-18

R-GAP: Recursive Gradient Attack on Privacy

Instead of searching for a match at all, R-GAP solves for the input directly, one layer at a time, from output back to input:

$$z_l = W_l\, a_{l-1}, \quad a_l = \sigma(z_l), \qquad \frac{\partial \ell}{\partial W_l} = \frac{\partial \ell}{\partial z_l}\, a_{l-1}^{\top}$$

- Each layer's equation is solved exactly by linear algebra – *no* optimization loop
- Recovers $a_0 = x$ exactly, subject to each layer's rank/invertibility condition

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DP-SGD as a Post-Hoc Defense

Reuses DP-SGD's clip-and-noise mechanics against exactly this leakage – applied once to the already-aggregated gradient, not the per-example, training-integrated, formally-private mechanism of DP-SGD itself.

$$\tilde{g} = \mathrm{clip}(g, C) + \mathcal{N}\big(0,\, (\sigma C)^2 I\big), \qquad \mathrm{clip}(g, C) = g \cdot \min\left(1, \frac{C}{\|g\|_2}\right)$$

- Clip to global L_2 norm C, then add Gaussian noise of multiplier σ
- Same (C, σ) grid swept against every attack, to compare how much noise each one tolerates

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Evaluation: Four Research Questions

Each RQ primarily varies one factor relative to a shared anchor (LeNet, CIFAR-10, batch 1, no defense):

- **RQ1.** Same architecture, dataset, and batch size – how do an optimization-based attack (DLG/iDLG) and a closed-form attack (R-GAP) compare in reconstruction quality and runtime?
- **RQ2.** Inverting Gradients is designed to scale – does it still succeed when the victim architecture is deeper and closer to practice (ResNet-18)?
- **RQ3.** How does each attack's reconstruction quality change when the leaked gradient is averaged over a batch of two private images instead of one?
- **RQ4.** Can DP-SGD's clipping and noise suppress each attack, and does the noise level needed to do so differ between attack families?

Evaluation: Victim Configurations

Every configuration evaluated across the four RQs:

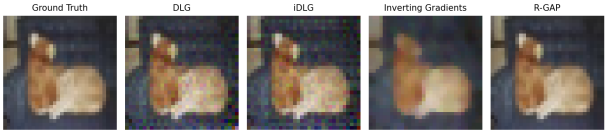
Method	Architecture	Dataset	Batch sizes	DP-SGD swept
DLG / iDLG	LeNet (bias, sigmoid)	CIFAR-10	1, 2	Yes
Inverting Gradients	ResNet-18 (untrained)	ImageNet (sample images)	1, 2	Yes
Inverting Gradients	LeNet (shared with DLG/iDLG)	CIFAR-10	1	Yes
R-GAP	LeNetRGAP (bias-free, LeakyReLU)	CIFAR-10	1, 2	Yes (batch 1 only)
R-GAP	FCN3 (bias-free, fully-connected)	CIFAR-10 (resized)	2	No

- The LeNet row shared by DLG/iDLG and Inverting Gradients uses identical architecture, initial weights, and image – giving RQ1 and RQ4 a matched-architecture comparison point across the two attack families

RQ1: Baseline Comparison, Shared LeNet

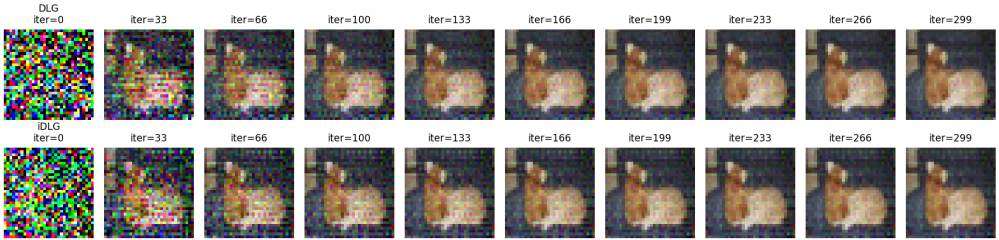
All four methods, same LeNet, same image, batch size 1, no defense:

Method	MSE	PSNR
DLG	0.00140	28.55 dB
iDLG	0.00160	27.95 dB
Inv. Gradients	0.0022	26.48 dB
R-GAP	$< 10^{-7}$	∞ (exact)



RQ1: Convergence of DLG vs. iDLG

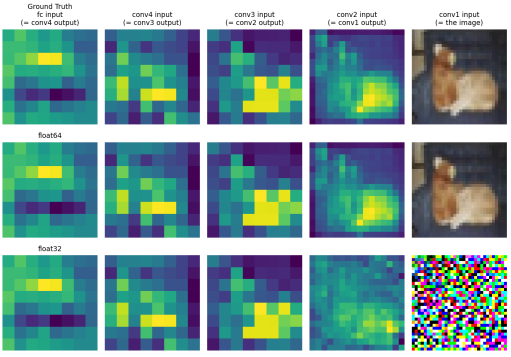
Both start from the same random noise; this is how they reach the result above.



- 10 milestones over the same 300-iteration L-BFGS run
- iDLG's single-variable search settles into shape earlier than DLG's joint search

RQ1: R-GAP Needs Float64

R-GAP's exactness above is conditional on precision – repeating it in float32 fails:



- float64: exact (MSE $< 10^{-7}$) vs. float32: MSE $= 91.25$, PSNR $= -19.6$ dB
- Failure localized to the final linear solve, where the smallest singular value amplifies rounding error

RQ2: Scaling to ResNet-18

Does Inverting Gradients' design actually deliver at a realistic scale?



- Untrained ResNet-18, 11.7M params, 224×224 image – an order of magnitude larger than RQ1's LeNet
- MSE = 0.0185, PSNR = 17.33 dB – visibly identifiable, but well below RQ1's 26.48 dB on the smaller network

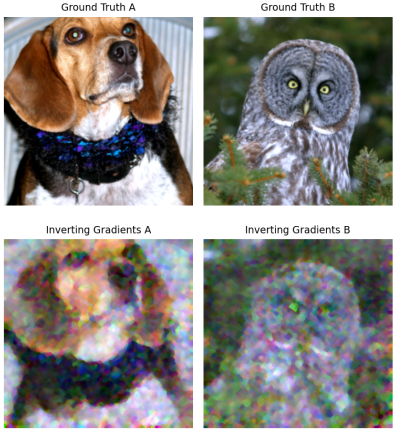
RQ3: Batch Size 1 vs. 2 – Optimization-Based Attacks

Each attack is compared against its own batch-size-1 result, when the leaked gradient instead averages two private images:

Method (architecture)	Batch 1 (PSNR)	Batch 2 (mean PSNR)
DLG (LeNet)	28.55 dB	16.29 dB
iDLG (LeNet)	27.95 dB	16.65 dB
Inverting Gradients (ResNet-18)	17.33 dB	16.35 dB

- Both LeNet attacks lose roughly 11–12 dB going from batch 1 to batch 2
- Inverting Gradients on ResNet-18 loses under 1 dB – its batch-1 result was already far from perfect (RQ2), leaving less margin left to lose

RQ3: Inverting Gradients, ResNet-18, Batch Size 2



- Both batch-2 reconstructions remain individually identifiable, despite the small PSNR drop from batch 1

RQ3: R-GAP at Batch Size 2 – the Linear-Combination Property

With $B = 2$ the leaked gradient averages both images – does R-GAP's exact solve degrade the same way on every architecture?

Architecture	Cosine similarity to ground truth vs. image A	vs. image B	vs. average $(A+B)/2$
LeNetRGAP (CNN)	0.062	0.055	0.061
FCN3 (fully-connected, LeakyReLU)	0.959	0.967	1.000

- On FCN3 (no conv layers), R-GAP's recursive solution closely matches the *direction* of the average of the two inputs – R-GAP inverts its LeakyReLU exactly, so nonlinearity isn't what matters here
- A convolutional architecture (LeNetRGAP) adds extra forward-pass constraints that this blended, two-image signal does not satisfy – the reconstruction matches neither image nor their average

RQ3: R-GAP Batch-2 Reconstructions



- LeNetRGAP/CNN (top): visibly uncorrelated noise
- FCN3 (bottom): closely matches the direction of the pixel average

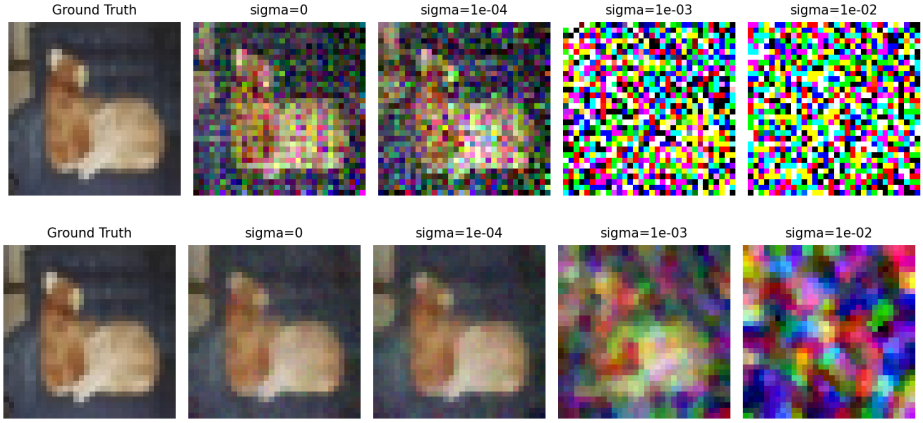
RQ4: DP-SGD vs. DLG Family (LeNet)

Same (C, σ) grid, same LeNet – does clipping/noise hit the L_2 and cosine cost functions equally hard?

Method (architecture)	PSNR (dB) at $C = 1.0\|g\|$ (no clipping) $\sigma = 0$	10^{-4}	10^{-3}	10^{-2}
iDLG (LeNet)	16.14	14.05	−8.11	−18.83
Inverting Gradients (LeNet)	27.94	27.32	17.34	10.75
Inverting Gradients (ResNet-18)	12.72	12.17	9.71	9.43

- Clipping alone already separates the two costs: iDLG's L_2 cost is sensitive to the gradient's magnitude and collapses under aggressive clipping even at $\sigma = 0$; Inverting Gradients' cosine cost is scale-invariant and barely changes
- Increasing σ then degrades both, but iDLG collapses sharply between 10^{-4} and 10^{-3}, while Inverting Gradients stays above 17 dB at that point

RQ4: Reconstructions Under DP-SGD (LeNet)



- iDLG (top) turns to pure noise well before Inverting Gradients (bottom) does, at the same noise levels

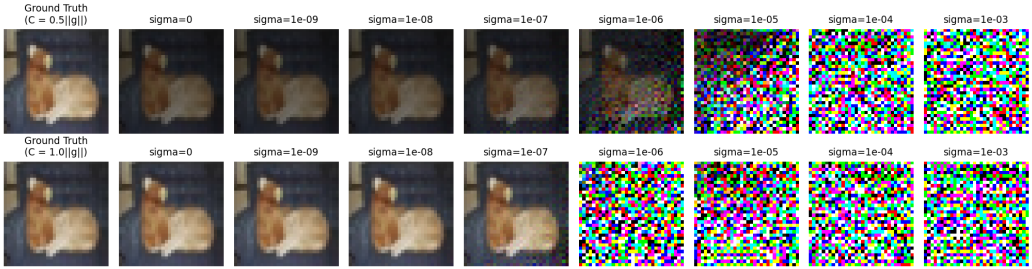
RQ4: DP-SGD vs. R-GAP

R-GAP needs a far finer noise grid than the attacks above – the same grid would overshoot its breaking point entirely and show nothing but collapse:

	PSNR (dB) at $C = 1.0\|g\|$ (no clipping)					
σ	0	10^{-9}	10^{-8}	10^{-7}	10^{-6}	10^{-5}
R-GAP (LeNetRGAP)	∞	73.2	51.6	31.0	-19.5	-18.5

- Exact at $\sigma = 0$, as in RQ1; still 31.0 dB at $\sigma = 10^{-7}$
- Collapses to -19.5 dB at $\sigma = 10^{-6}$ – just one order of magnitude more noise, three to four orders of magnitude below where iDLG breaks

RQ4: R-GAP Reconstructions Under DP-SGD



- The break between $\sigma = 10^{-7}$ and 10^{-6} is visible by eye, at a noise level too small to touch DLG, iDLG, or Inverting Gradients at all

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Key Takeaways

- Gradients alone are enough to reconstruct private images/labels
- Closed-form (R-GAP) is exact and fastest, but fragile: precision and batch size
- Optimization-based attacks scale further but converge less precisely
- DP-SGD noise that defeats R-GAP leaves DLG/iDLG visibly unaffected
- No single defense budget suppresses every attack family equally

Thank You

Questions?